Domain Name System
DNS

Understanding DNS Name Resolution

- IP addresses - formed from four octets of numbers from 1 to 254
  - 192.168.1.21 for example
- Name resolution is the process of converting these numerical IDs into names the average person can read and remember
  - www.google.com for example

Understanding DNS Name Resolution

- DNS created a hierarchical namespace
  - Root (.)
  - Top-level domain (.edu, .gov, .ca)
  - Second-level domain (pbcc, google, military)
  - Host (www, tcpip1, ftp)
- FQDNs – www.pbcc.edu
- Relative distinguished names - www

Domains

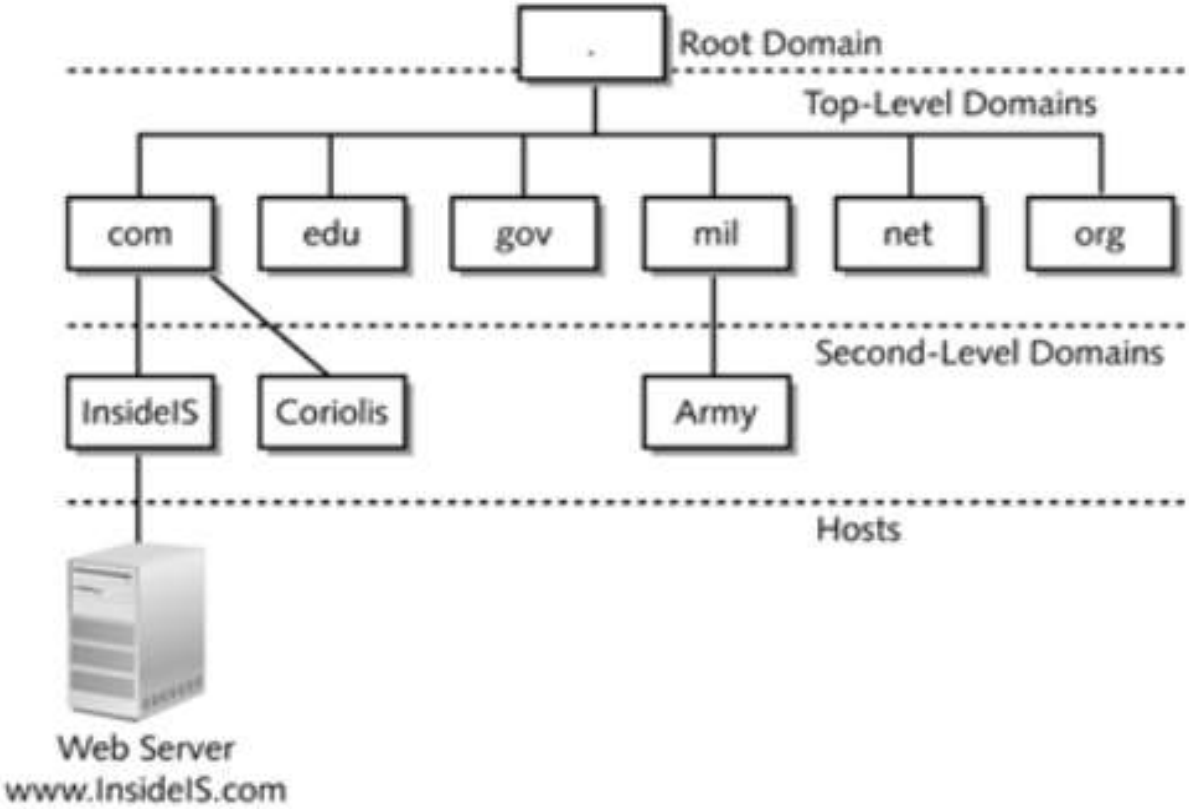


Figure 4-1 An example of the domain namespace

Understanding DNS Name Resolution

- Forward lookup query - standard method of name resolution in DNS
- Client sends a query to its primary DNS server, which in turn forward the request (if necessary) until the name is fully resolved
- A records

Understanding DNS Name Resolution

- Sites are then cached on the primary DNS server, to avoid unnecessary traffic
- Reverse lookup queries can be used in troubleshooting, or for security purposes
- PTR records

Operation of DNS

- Walk the tree
  - Progressive query the DNS database structured from the root (.) to top level (com) to second level (google) to subdomain (business) to www.

Understanding the Different Types of Name Servers

- Primary DNS servers
- Secondary DNS servers
- Caching-only name servers
- Forwarding DNS servers

Primary DNS Servers

- The primary DNS server is the authoritative server for the zone
- Any changes to the DNS domain must be made here
- Read/Write Copy of DNS Database

Secondary DNS Servers

- Serves as a backup for the primary DNS server (failover)
- Can also be used for load balancing
- Read Only Copy of DNS Database

Caching-only Name Servers

- Functions only to cache name resolution queries
- Advantage is that network traffic is significantly reduced

Monitoring and Troubleshooting DNS for AD

- Name resolution is first attempted through cache.
- Clear local names that have been cached.
- ipconfig /flushdns

Troubleshooting DNS for AD

- Use **nslookup** for troubleshooting and as a security tool
  - server tcpip1.central.pbcc.edu – connect to DNS server you are testing
  - tcpip4.north.pbcc.edu – attempt to resolve a name to ip address
- **ping** FQDN – ping tcpip4.north.pbcc.edu

DNS

- Data transmission in IP networks needs actual numeric IP addresses
- People want to use names – mnemonic, memorable names
- Need a database of name-to-IP-address mappings
  - Allow people to use textual type names
- Need a way that programs can access this database
- DNS evolved to meet these needs

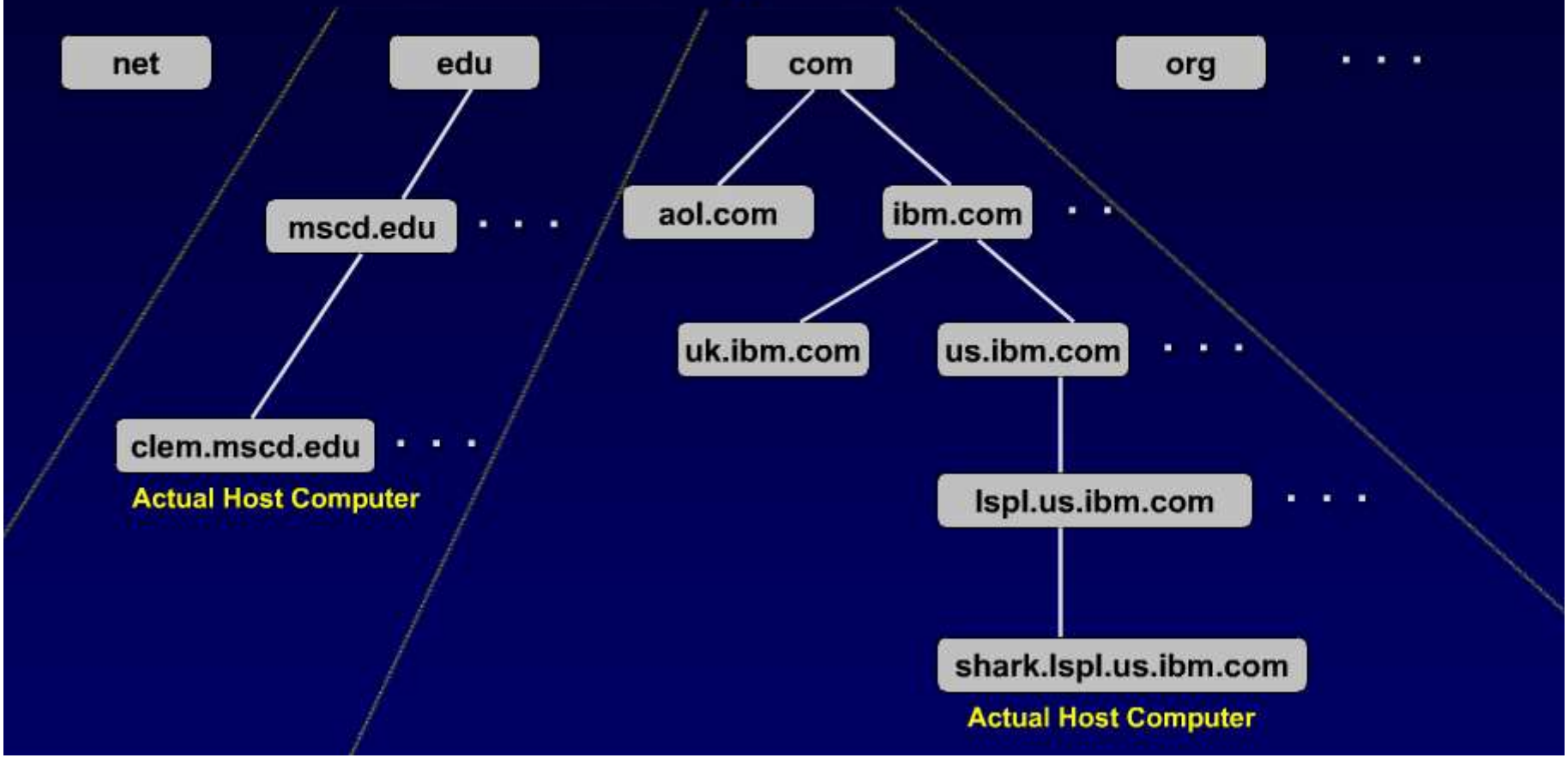
Elements of DNS

- Hierarchical Naming System and conventions
- Naming data kept in DNS servers in a massively distributed database of all name-address mappings
  - DNS Servers needed – break up "authority" by domain name suffixes and zones
  - Each DNS server keeps a database – a set of records – containing the name-IP-address "bindings"
- Resolver software needed on hosts
  - Each host (and its resolver) needs to know about a default or local DNS server
- Messaging protocol to request records and send responses
  - Each time host has to resolve a name, its resolver sends a request to the local DNS server, which ultimately returns an authoritative response (We won't cover layout of these messages)

Hierarchical Naming System

- "Labels" (made of letters and numbers) separated by dots (lspl.us.ibm.com, etc.)
- Hierarchical, most general on right, most specific on left
- Number of dots not specified
  - Only top-level domains (TLDs) specified
  - Each organization can choose how to give out more specific names, how many levels
- Rightmost is TLD; Leftmost is computer name
- Name Space is first broken into top level domain suffixes (.com, .edu, etc.), then each of these is broken into domain names (ibm.com, aol.com, etc.), then each is broken into sub-domains, etc.

Hierarchical Naming – Division of Name Space



Top-Level Domains

Domain Name	Assigned To
aero	Air transport industry
arpa	Infrastructure domain
biz	Businesses
com	Commercial organization
coop	Cooperative associations
edu	Educational institution
gov	United States Government
info	Information
int	International treaty organizations
mil	United States military
museum	Museums
name	Individuals
net	Major network support center
org	Non-commercial organizations
pro	Credentialed professionals
country code	A country

Top-Level Domains

- Internet Corporation for Assigned Names and Numbers (ICANN) – how to register for a domain
  - InterNIC administers domain name registration for ICANN
  - See: http://www.internic.net/ (see **Registry Whois**)
- Also: geographic names – are considered Top Level Domains
  - .co.us
  - .ac.uk for academic in Britain
- For latest list of current TLD names and who owns them (registrars), see http://www.iana.org/domain-names.htm

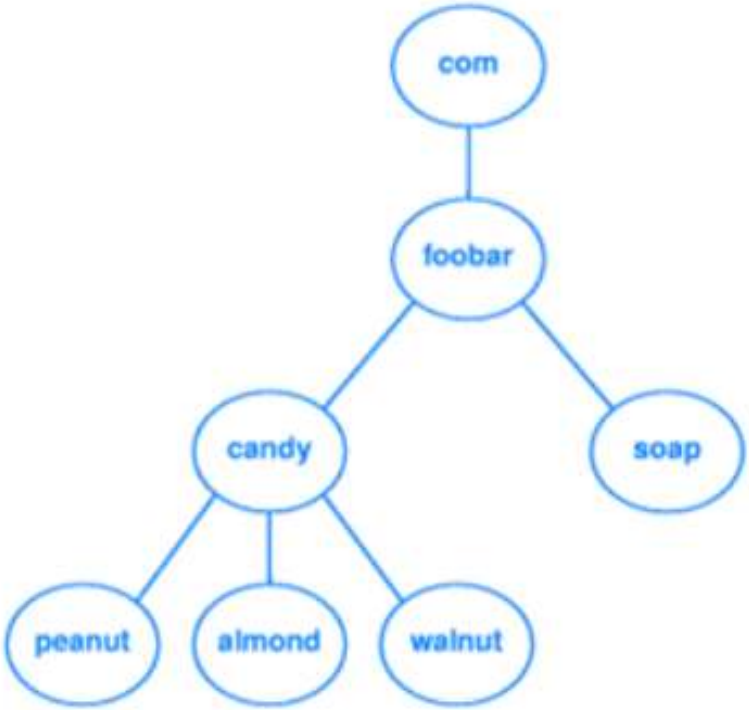


Figure 29.2 A graphical representation that illustrates one way a DNS hierarchy might be structured in a corporation. Names for individual computers can be added to the diagram as well.

Data of DNS Database

- Resource Records
- Contain <name, IP-address> pairs
- Resource records contained in servers scattered across internet
- Software on requesting host becomes client of these “DNS servers”

DNS Name Database Broken Up Hierarchically Among Domain Name Servers

- At the highest level we have a few top level Domain Servers in the world – *root servers* (~13)
- Within each TLD, next level domain servers (like for ibm.com) provide authority for all naming for anything with its suffix (owns all ibm.com)
- Each registered organization has autonomy from its point down in the hierarchy (ibm.com)
  - Controls its prefix
  - Can define nested domains inside theirs
- Each server has authority for a part of the hierarchy; set of machines managed by a name server is called a zone – zones usually nested

Name Database

- Root servers do not contain mappings for all machines, only for TLD servers
- They refer request to DNS servers farther down the hierarchy
- Requests are handled iteratively and recursively

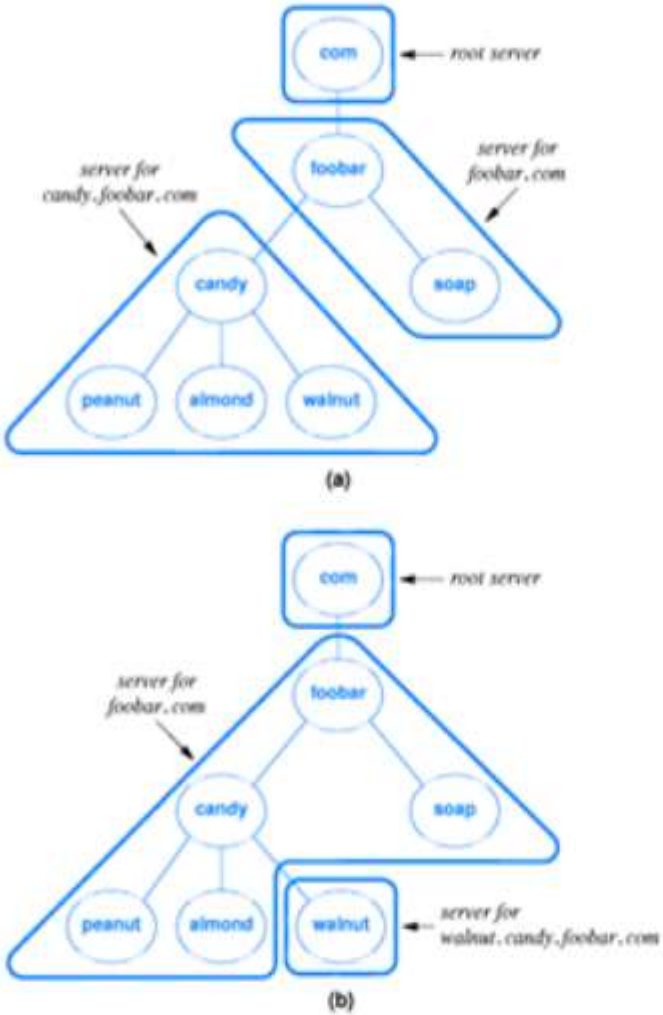
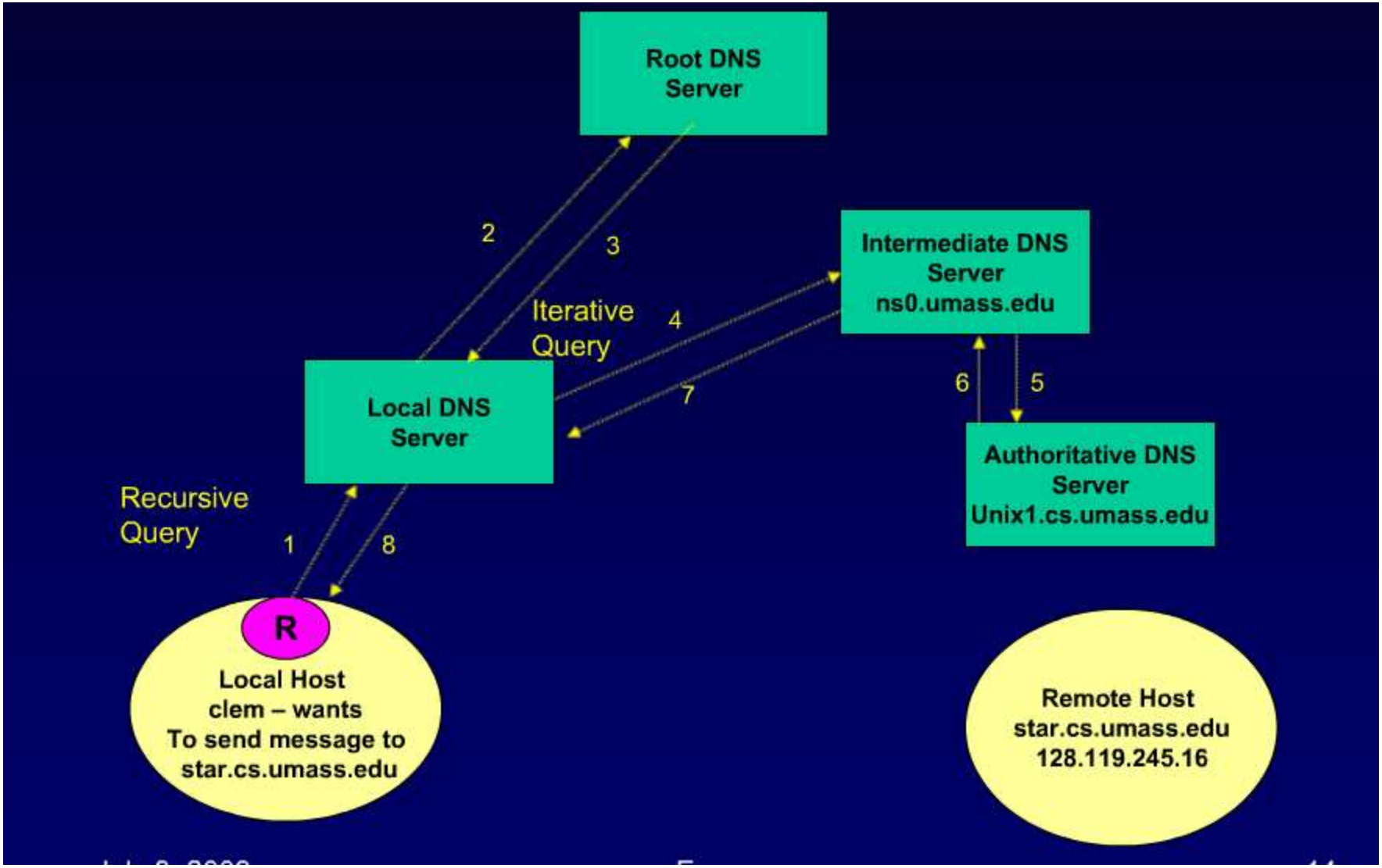


Figure 29.3 Two examples of ways the domain name hierarchy from Figure 29.2 can be divided among three servers. Each organization chooses how to divide names among its servers.

Example Name Resolution



Name Resolution – Role of the Local Resolver

- Translation from name to IP or back is handled by *resolver* software in local host
  - Unix has: gethostbyname; Windows resolver
- Resolver is part of operating system: Accessed via system call
- Resolver acts as client to local DNS server
- Knows local DNS server, sends request to it
- If local server does not know answer (not authority for domain), local server sends request to root level
  - Server becomes client
- Resolver can also handle abbreviations ("clem" instead of clem.mscd.edu) locally

Name Resolution

- Process repeats until mapping found – iterative or recursive request
- Host resolver first goes to local Name Server – iterative example
  - Local NS (LNS) handles request and replies if authoritative for the given name
  - Otherwise sends to root server for TLD
  - Root server replies to LNS with address of NS (NS1) that handles target domain
  - Now LNS sends request to NS1
  - Either NS1 gives response or passes LNS address of another NS (NS2) – in which case LNS sends request to NS2. Etc.
  - Eventually LNS get authoritative response
  - LNS sends final response to host resolver

Performance Considerations

- UDP typically used instead of TCP
- Because of very heavy traffic, root servers are replicated
- Zone structure depends upon spatial locality of reference - Users tend to look up local computers more often
- Look up addresses from same domain name over and over
- Locality of time - Mappings *cached* to speed lookups – local DNS servers cache what they learn (with time limits)
- A good story can be found at: http://www.spirit.com/Network/net1102.html

Some RR Types Kept in DNS Servers

RR Type	Explanation, Format
A	Name → Address
MX	Domain name → email exchanger
CNAME	Alias name → canonical name
PTR	Reverse lookup (in-addr.arpa → name)
SOA	Start of Authority: one SOA record in each zone – gives domain name, responsible party, etc
NS	Authoritative Name Server

Notes on Resource Records

- RRs have TTL field – don't live forever in cache
- Might not be able to 'ping' mail (or other) machine – you only get the type of records you ask for! There may be an MX record for a computer but no A record!!
- The id-addr.arpa Domain – faster reverse lookups
  - Reverse record also stored by some DNS servers (IP address backwards!)
  - Compare A-type record:
  - Name: star.cs.umass.edu Address: 128.119.245.16
  - To the PTR – type record
  - 16.245.119.128.in-addr.arpa name = star.cs.umass.edu

Examples of DNS Records

```
IN  SOA clem.mscd.edu. root.clem.mscd.edu. ( 954
  10800 3600 604800 86400 )
IN  NS  clem.mscd.edu.
IN  NS  thor.mscd.edu.
mscd.edu.        IN  MX   10   clem.mscd.edu.
localhost        IN  A    127.0.0.1
kbsl022          IN  A    147.153.245.23
9cad438          IN  A    147.153.228.186
plad582          IN  A    147.153.179.7
```

Use nslookup To Get Name Resolutions and Root Servers

- Nslookup interactive mode
- > host
- > set type=[a | ns | mx | ptr | cname]
- > domain
- > . to show root servers
- > ? (help)
- http://clem.mscd.edu/~evansell/NETWORKS/nslookupUnix.html
- One example of many online NS LOOKUP tools is at: http://www.nslookup.be/index.jsp